

A Proposed Lower Bound for Erdős Problem 119

Samuel Korsky

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Let $(z_j)_{j \geq 1}$ be a sequence on the unit circle, and define

$$p_k(z) := \prod_{j=1}^k (z - z_j), \quad M_k := \max_{|z|=1} |p_k(z)|.$$

After some discussion with GPT 5.6-Pro about Leja sequences inspired by [arXiv:2006.10708](#), and the guiding idea that if M_t is small, then the points $z_1, \dots, z_t$ must be quantitatively close to equidistributed at low frequencies, GPT 5.6-Pro claims the lower bound (which I plan to review extremely carefully, especially given that the problem has a monetary value)

$$\sum_{k \leq N} M_k \gg \frac{N^{5/4}}{\sqrt{\log N}}.$$

Write $z_j = e^{2\pi i x_j}$ and define

$$B(x) := \log |2 \sin \pi x|, \quad F_k(x) := \sum_{j \leq k} B(x - x_j), \quad L_k := \log M_k = \max_x F_k(x).$$

Here $x_j \in \mathbb{R}/\mathbb{Z}$, integrals are taken over $\mathbb{R}/\mathbb{Z}$ with normalized Lebesgue measure, and logarithms are natural.

I found no critical error after a brief review, so wanted to post the ideas here. The proof has the following steps.

1. For

$$S_m(t) := \sum_{j \leq t} e^{-2\pi i m x_j},$$

the function $L_t - F_t$ is nonnegative, has integral L_t , and has m th Fourier coefficient $S_m(t)/(2m)$ for $m \geq 1$. Hence

$$|S_m(t)| \leq 2mL_t.$$

2. Let K_H be the Fejér kernel and let $B_H := K_H * B$. Since $K_H \geq 0$ and has integral 1,

$$L_k \geq (K_H * F_k)(x_{k+1}) = \sum_{i \leq k} B_H(x_{k+1} - x_i).$$

3. Summing over $k < t$ counts every unordered pair exactly once:

$$\sum_{k < t} L_k \geq \sum_{1 \leq i < j \leq t} B_H(x_j - x_i).$$

4. Let

$$\mathcal{H}_H := \sum_{i \leq H} \frac{1}{i}.$$

The Fourier expansion

$$B_H(x) = - \sum_{m=1}^H \left(1 - \frac{m}{H+1}\right) \frac{\cos(2\pi m x)}{m}$$

and the identity

$$|S_m(t)|^2 = t + 2 \sum_{i < j} \cos(2\pi m(x_j - x_i))$$

give, using Step 1,

$$\sum_{k < t} L_k + \frac{H(H+2)}{3} \cdot L_t^2 \geq \frac{t}{2} \left(\mathcal{H}_H - \frac{H}{H+1} \right).$$

5. Take $t = N$. If $L_N > 2 \log N$, then $M_N > N^2$, so the result is immediate. Otherwise choose

$$H = \left\lfloor \frac{\sqrt{N}}{\log N} \right\rfloor.$$

Then Step 4 gives

$$\sum_{k < N} L_k \geq \frac{N}{4} \cdot \log N - \frac{N}{2} \cdot \log \log N - O(N).$$

By AM-GM,

$$\sum_{k < N} M_k = \sum_{k < N} e^{L_k} \geq (N-1) \exp \left(\frac{1}{N-1} \sum_{k < N} L_k \right) \gg \frac{N^{5/4}}{\sqrt{\log N}}.$$